

Profile

Aspiring Electrical Engineer driven by curiosity and a passion for technical challenges. Currently seeking a stimulating 6-month End-of-Studies Internship (Capstone Project), I aim to leverage my adaptability and eagerness to learn to contribute to your innovative projects.

Education

- 2023 – 2026 **Engineering Degree, ENSET, Mohammedia**
Major: Electrical Engineering and Industrial Systems Control (GECSI).
- 2022 – 2023 **Bachelor in Physical Sciences, Faculty of Sciences Ben Msik, Casablanca**
(Electronics)
- 2020 – 2022 **Associate Degree in Physics, Faculty of Sciences Ben Msik, Casablanca**
(DEUG)
- 2019 – 2020 **Technical Baccalaureate, Technical High School, Mohammedia**
Electrical Option.

Professional Experience

- Jul 2025 – Sep 2025 **Engineering Internship, SMCV, Mohammedia**
Focus: Automation & Design
- Programming and supervision of a plastic injection machine.
 - Design and integration of a mechanical arm for nut insertion process optimization.
 - Complete study, design, and automation of a hot nut insertion press.
- Jul 2024 – Aug 2024 **Technical Internship, ONEE, Mohammedia**
Focus: Electrical Grid
- Analysis of Siemens digital relays and development of testing procedures.

Technical Projects

- Electronics Reverse Engineering, Simulation, and Optimization of a Phase-Controlled Triac Dimmer for a 1.4 kW Universal Motor.
Design and simulation: Class D Amplifier, Single-phase SPWM Inverter, Regulated Boost Converter, and Filtered Rectifier.
- Embedded Sys. Control and supervision of a pump using ESP32 and web interface.
Design and PCB implementation of an L-C Meter.
- Automation Programming and supervision of a Smart Conveyor (PLCnext PLC, Schneider VFD).
Reprogramming of a mattress spring assembly machine (Delta PLC).
- Modeling Design of an orbital autopilot (Luenberger Observer + State Feedback) for stabilization.
Non-linear control (Backstepping Law) of a Micro-Grid wind turbine.
Output feedback control for fast charging (CC-CV) of Li-ion batteries.
- Simulation Finite Element Analysis of TE10 mode propagation (Waveguide).

Technical Skills

- Expertise Industrial Automation & Supervision, Power Electronics & Electrical Grids, Embedded Systems & IoT, Modeling & Advanced Control.
- Software Siemens TIA Portal, PLCnext Engineer, MATLAB/Simulink, Caneco BT, ETAP, Proteus, PyTorch, Google Colab, Ansys (HFSS/Mechanical), Autodesk Inventor, AutoCAD.
- Programming C/C++, Python, Ladder/SCL, VHDL.
- Languages **Arabic** (Native), **English** (C2 - Advanced), **French** (B2 - Intermediate).